

PRODUCT SHEET

HYDROSENSE PRO ULTRA LEGIONELLA WATER TEST

This kit is designed to test for Legionella in risk areas such as:

- » Domestic and industrial hot and cold water systems.
- » Cooling towers.
- » Decorative fountains, hot tubs and pools.
- » Sinks and showers.
- » Mistlers, sprinklers, air washers, humidifiers and more.



Hydrosense PRO Ultra Legionella Water Test Product Code 500500

The test contains all the items required to perform 1 test.

- » The test detects *Legionella pneumophila* serogroups 1-15 (LpSG1-15).
- » Using the hollow fibre filter technology included in the kit, the sensitivity of the test is 100 cfu/L (0.1cfu/ml) for Lp sg1 and 100 cfu/L (0.1cfu/ml) for Lp sg2-15.
- » Test detects sg1 on one test line and sg2-15 on a separate test line.
- » The test detects Viable But Non Culturable (VBNC) and *Legionella pneumophila* sg1-15 in all stages of its life cycle.
- » The test is simple to use, no interpretation is required.
- » No need to measure or dilute solutions.
- » Additional action levels of 1000 cfu/L and 10,000cfu/L provided using the included scorecard for a better understanding of the risk.
- » Provides straightforward results, on-site, within 25–35 minutes.
- » The test helps to minimise delays and downtime.
- » Regular use increases confidence in Legionella controls.
- » Test results can be recorded via a form download from www.hydrosense-legionella.com for greater traceability of testing and results.

Hydrosense uses Lateral Flow Immunochromatographic Assay (LFICA) technology to detect the presence of cell surface antigens from Legionella bacteria. It uses Legionella antibodies, tagged with coloured nanoparticles, which bind to any bacteria found in the sample and make them visible on a line on the device. The test works just like a pregnancy test – one line indicates a negative result and two or three lines indicate a positive test.

PRO Results Reference

Negative



Positive sg1



Positive sg2-15



Positive sg1-15



Bacteria detected *Legionella pneumophila* serogroups 1-15

Detection method Lateral flow immunochromatographic assay

Lower limit of detection 100 cfu/L (0.1cfu/ml) *L. pneumophila* sg1 and 100 cfu/L (0.1cfu/ml) *L. pneumophila* sg2-15

Specificity

Tested against a wide range of waterborne bacteria, including *Acinetobacter calcoaceticus*, *Aeromonas hydrophila* subsp. *hydrophila*, *Bacillus subtilis*, *Burkholderia cepacia*, *Citrobacter freundii*, *Citrobacter koseri*, *Escherichia coli*, *Enterobacter cloacae*, *Klebsiella oxytoca*, *Pseudomonas aeruginosa*, *Pseudomonas fluorescens*, *Pseudomonas putida*, *Pseudomonas stutzeri*, *Ralstonia pickettii*, *Raoultella terrigena*, *Streptococcus pyogenes*, *Yersinia ruckeri*.

Weak cross reactions can occur on either test line if high concentrations of the opposing serogroups are present. See table below for levels. These levels indicate very high levels of contamination and would be a cause for action to be taken. *Staphylococcus aureus* has been observed to cross-react with the test at concentrations > 10E08 CFU/L, which is higher than found in most water samples.

Organism	≥CFU/ml
<i>L. pneumophila</i> sg1	1x10E07
<i>L. pneumophila</i> sg- 2, 3, 8, 11, 13, 14	1x10E08
<i>L. pneumophila</i> sg- 4, 5, 6, 7, 9, 10, 15	1x10E07
<i>L. pneumophila</i> sg12	8x10E06
<i>S. aureus</i>	2x10E08

Test volume 100 µL

Suggested sample volume 250 mL

Sample parameters Sample should filter in less than 10 minutes.

Biocides

The product has been tested with a range of common biocides and biodispersants used in *Legionella* control, at their normal maximum operating concentration. Active ingredients tested include: glutaraldehyde, dodecyl-dimethyl-ammonium chloride, isothiazoles, dibromonitripropionamide, ionic and non-ionic surfactants.

The Test gives a false positive if used with polymeric biguanide or THPS. It should not be used to test such systems.

Operator time 5–10 minutes

Test time 25 minutes

Shelf life 18 months from date of manufacture, when stored at room temperature (<30°C) in original packaging. All test strips are hermetically foil-wrapped and with a printed expiry date.